

goJTAG, a JTAG/ Boundary Scan Tool

Wide adoption of standardised IEEE 1149.x test methods has called for the independent and non-commercial platform-based tools for JTAG test principles. goJTAG is a JTAG/boundary scan tool which offers a clear illustration of the standard test principles. Read this article to get an insight of how to start working with it

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The well-established test technique boundary scan JTAG IEEE Std. 1149.x is a unique solution to many test requirements and provides information about the board without accessing the complete board. Therefore it is ideal for testing the complex boards which could not be tested otherwise due to lack of test access. Boundary scan systems have little hardware and their significant part is software.

goJTAG is a part of an open source project which provides a full package for a JTAG newcomer including training material, slides and exercises. The software allows you to study boundary scan techniques, carry out your own experiments, perform diagnosis and much more. The software environment comes with a built-in example project, and you can create your own projects with descriptions of boards, chips, cluster logic and so on.

goJTAG includes a simulation component fully illustrating the underlying concepts of JTAG/boundary scan that allows you to directly step-wise control the test features of IEEE 1149.1-compliant devices and observe a system's reaction in real time in an on-screen simulation environment.

Starting with goJTAG

After successful installation of the software, you can run it, which will

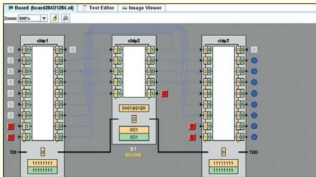


Fig 1: Board



Fig 2: Diagnostic results

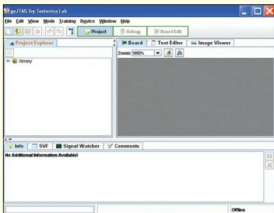


Fig 3: GUI

open a GUI environment providing you with various options to control the test features of IEEE 1149.1-compliant devices.

Working modes. There are three working modes supported by goJTAG:

1. Project Mode allows you to view/organise your project items, create or import new files

2. Using De-
bug Mode, you can

perform boundary scan operations such as select instructions, perform scan, define test vectors, observe diagnostic results, amongst others.

3. In Board Edit Mode, you can create/edit your own netlists.

For changing the mode, just select the appropriate mode from the dropdown Mode menu. Alternatively, you can select the corresponding special toolbar

buttons at the top.

Note: You should start your work with Project Mode to choose the netlist first.

Opening netlist. After creating a

System requirements

Runs on Windows, Unix and Linux
Note: Installation file can be found in EFY Plus DVD that came with this magazine.